

Leverage From Learning

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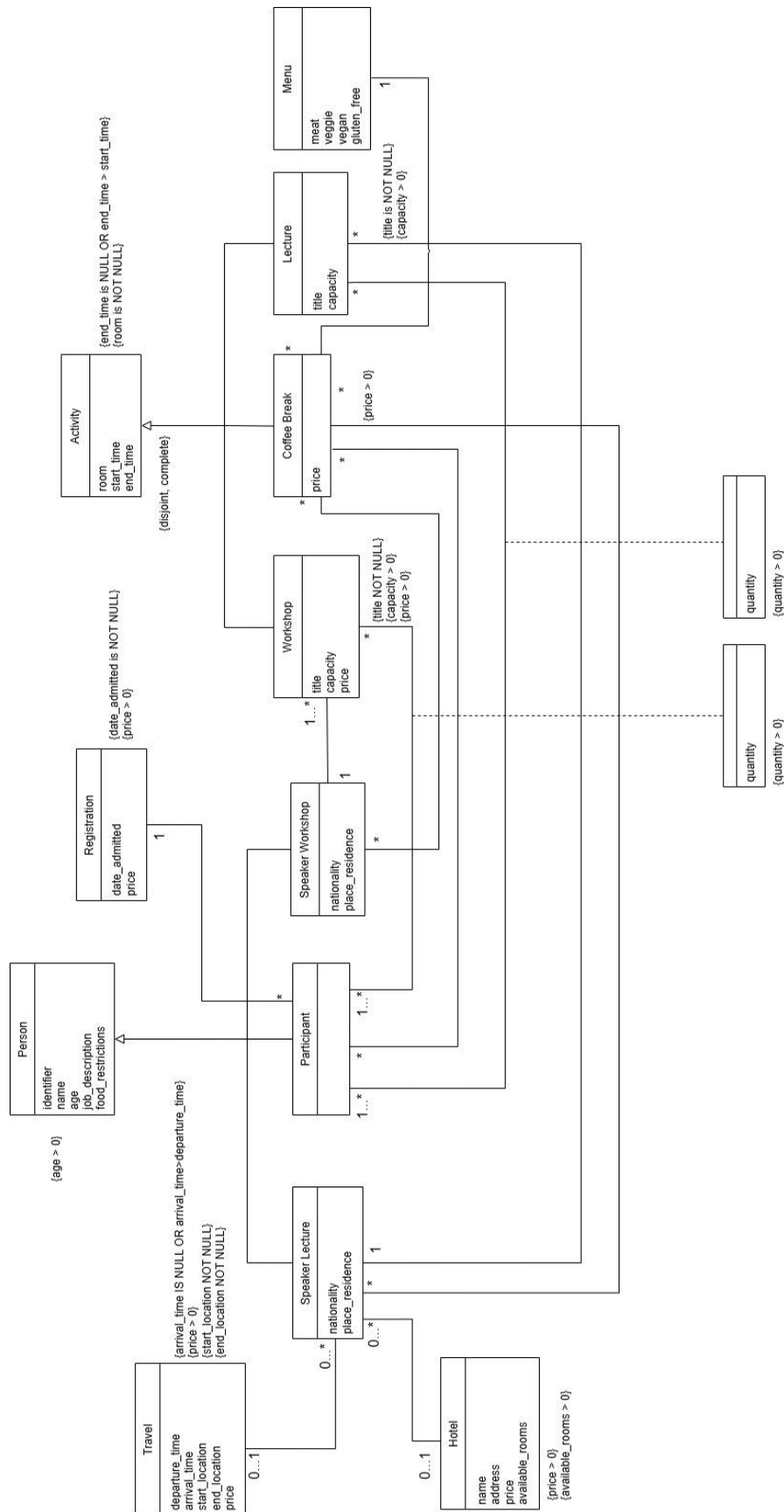
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Description:

Leverage From Learning (LFL) is a 1 day event organized by students dedicated for students and those who are interested in leveraging learning. This information system coordinates lectures, workshops and coffee breaks for all participants and speakers to attend.

- There are several speakers for lectures and workshops and participants. Each speaker and participant has an identifier number, name, age, job description, food restriction.
- Speakers also have a nationality and a place of residence associated.
- Participants must enroll in the event which has a certain registration date and price.
- Each lecture's speaker might need transportation. Transport has a departure and arrival time, start and end location, with its respective price.
- They might need to stay at a hotel, which has a name, address, number of available rooms and a price.
- The workshop speakers may be responsible for more than 1 workshop, with a title, capacity and price, but each workshop must have one speaker.
- Lecturers' speakers and participants can attend several activities: workshops, lectures and coffee breaks. Workshops' speakers can only go to coffee breaks and workshops.
- Each activity has a room, start time and end time. Workshops and lectures also have a title (theme of the event) and capacity. Workshops and coffee breaks have a price associated with the organization of the event.
- Coffee breaks' attendants have different menus at their disposal. During the event, there will be two coffee breaks.

UML Diagram:



Relational Model:

Person (identifier, name, age, job_description, food_restrictions)

CHECK(age > 0)

SpeakerLecture (identifier → Person, nationality, place_residence, travel → Travel, hotel → Hotel, coffee_break → CoffeeBreak)

Travel (travel_id, departure_time, arrival_time, start_location, end_location, price)

CHECK(arrival_time IS NULL OR arrival_time > departure_time)

CHECK(price > 0)

NOT NULL(start_location)

NOT NULL(end_location)

Hotel (reservation_id, name, address, price, available_rooms)

CHECK(price > 0)

CHECK(available_rooms > 0)

SpeakerWorkshop (identifier → Person, nationality, place_residence, coffee_break → CoffeeBreak)

Participant (identifier → Person, registration_id → Registration, coffee_break → CoffeeBreak)

Registration (registration_id, date_admitted, price)

NOT NULL(date_admitted)

CHECK(price > 0)

Activity (room, start_time, end_time)

CHECK(end_time IS NULL OR end_time > start_time)

NOT NULL(room)

Lecture (room → Activity, start_time → Activity, title, capacity, identifier → SpeakerLecture)

NOT NULL(title)

CHECK(capacity > 0)

CoffeeBreak (room → Activity, start_time → Activity, price, menu_id → Menu)

Workshop (room → Activity, start_time → Activity, title, capacity, price, identifier → SpeakerWorkshop)

Menu (menu_id, meat, veggie, vegan, gluten_free)

ParticipantWorkshop (participant → Participant, workshop → Workshop, quantity)

CHECK(quantity > 0)

ParticipantLecture (participant → Participant, lecture → Lecture, quantity)

CHECK(quantity > 0)